

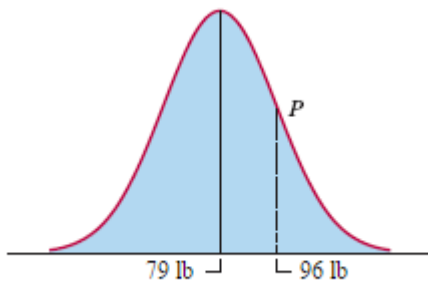
Quiz 2

Problem Solving, Math 1000 – Summer 2016

Friday, June 3

The following quiz is out of 40 points, and will count towards your quiz grade. Each problem is worth exactly 10 points. Answer the following problems (you may use your notes).

Problem 1 Consider the normal distribution shown below:



Find the mean, median, and standard deviation of the distribution.

Problem 2 Consider a normal distribution with first quartile $Q_1 = 62.275$ points and third quartile $Q_3 = 74.725$ points. Find the mean and standard deviation of the distribution.

Problem 3 Explain why a distribution with median $M = 82$, mean $\mu = 71$, and standard deviation $\sigma = 11$ cannot be a normal distribution.

Problem 4 A normal distribution has mean $\mu = 100$ points and first quartile $Q_1 = 93.25$ points. Find the z-value for $x = 85$ points.